

Exercise 1: Harmonics 1-16 (upper half of the string)

Play all of these notes by lightly placing the left-hand second finger at the pitch indicated by the diamond note-head. Play at a *mezzo forte* dynamic, taking care that the bow moves closer to the bridge as the notes get higher in pitch. All of these harmonics sound at the same pitch as indicated by the note-head.

The exercise is divided into four sections, each showing 16 positions on the string. The diamond note-head indicates the target pitch, and the arrow points to the left-hand second finger position. The A-String section is in treble clef, and the D-String, G-String, and C-String sections are in bass clef. A dashed line labeled '8va' indicates the octave range.

A-String

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

D-String

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

G-String

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

C-String

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Exercise 2: Harmonics 1-16 (all nodes)

Play the following in two ways:

- Begin by playing the last bar of each line – this is the last node of each harmonic, which sounds at pitch, as seen in Exercise 1. Take note of the bow placement and maintain it while playing the other nodes of the same harmonic.
- Play the last bars of each line sequentially (upper half of the string), followed by the first bars of each line (lower half of the string), in order to feel the increasingly small distances between the nodes. This is an important aspect of understanding the geography of the harmonics on the string. The first nodes of the 10th-15th harmonics have very slight deviations in finger placement, by less than an eighth-tone. These have been indicated in cents which in this case, as with the arrows, serve only as approximate indication of finger placement. Special attention should be given to aural recognition of pitch placement when familiarizing oneself with the upper harmonic, as these will not correspond to the notes of the diatonic scale.

A-String

The image displays musical notation for the A-string harmonics, specifically the 2nd through 9th harmonics. Each harmonic is represented by a staff with a treble clef and a key signature of one sharp (F#). The notation shows the positions of the nodes (indicated by dots) and the corresponding fingerings (indicated by numbers 1 through 8). The 2nd harmonic is shown with a single node at the midpoint (1/2). The 3rd harmonic has two nodes at 1/3 and 2/3. The 4th harmonic has two nodes at 1/4 and 3/4. The 5th harmonic has four nodes at 1/5, 2/5, 3/5, and 4/5. The 6th harmonic has two nodes at 1/6 and 5/6. The 7th harmonic has six nodes at 1/7, 2/7, 3/7, 4/7, 5/7, and 6/7. The 8th harmonic has four nodes at 1/8, 3/8, 5/8, and 7/8. The 9th harmonic has eight nodes at 1/9, 2/9, 4/9, 5/9, 7/9, and 8/9. The notation also includes arrows indicating the direction of the bow stroke and the placement of the fingers.

The image displays musical notation for the 10th through 16th harmonics of a string, likely a guitar. Each harmonic is represented by a staff with a treble clef and a key signature of one sharp (F#). The notation includes fret positions indicated by arrows and the corresponding frequency ratios.

10th harmonic: Fret positions 1/10, 3/10, 7/10, 9/10. Ratios: 1/10, 3/10, 7/10, 9/10.

11th harmonic: Fret positions 1/11, 2/11, 3/11, 4/11, 5/11, 6/11, 7/11, 8/11, 9/11, 10/11. Ratios: 1/11, 2/11, 3/11, 4/11, 5/11, 6/11, 7/11, 8/11, 9/11, 10/11.

12th harmonic: Fret positions 1/12, 5/12, 7/12, 11/12. Ratios: 1/12, 5/12, 7/12, 11/12.

13th harmonic: Fret positions 1/13, 2/13, 3/13, 4/13, 5/13, 6/13, 7/13, 8/13, 9/13, 10/13, 11/13, 12/13. Ratios: 1/13, 2/13, 3/13, 4/13, 5/13, 6/13, 7/13, 8/13, 9/13, 10/13, 11/13, 12/13.

14th harmonic: Fret positions 1/14, 3/14, 5/14, 9/14, 11/14, 13/14. Ratios: 1/14, 3/14, 5/14, 9/14, 11/14, 13/14.

15th harmonic: Fret positions 1/15, 2/15, 4/15, 7/15, 8/15, 11/15, 13/15, 14/15. Ratios: 1/15, 2/15, 4/15, 7/15, 8/15, 11/15, 13/15, 14/15.

16th harmonic: Fret positions 1/16, 3/16, 5/16, 7/16, 9/16, 11/16, 13/16, 15/16. Ratios: 1/16, 3/16, 5/16, 7/16, 9/16, 11/16, 13/16, 15/16.